

Class - 07

DENSITY CURVE

Data Distribution

Density Curve

Let's understand about **Density Curve**,

A **density curve** is a smooth curve that shows the **overall shape of a data distribution**.

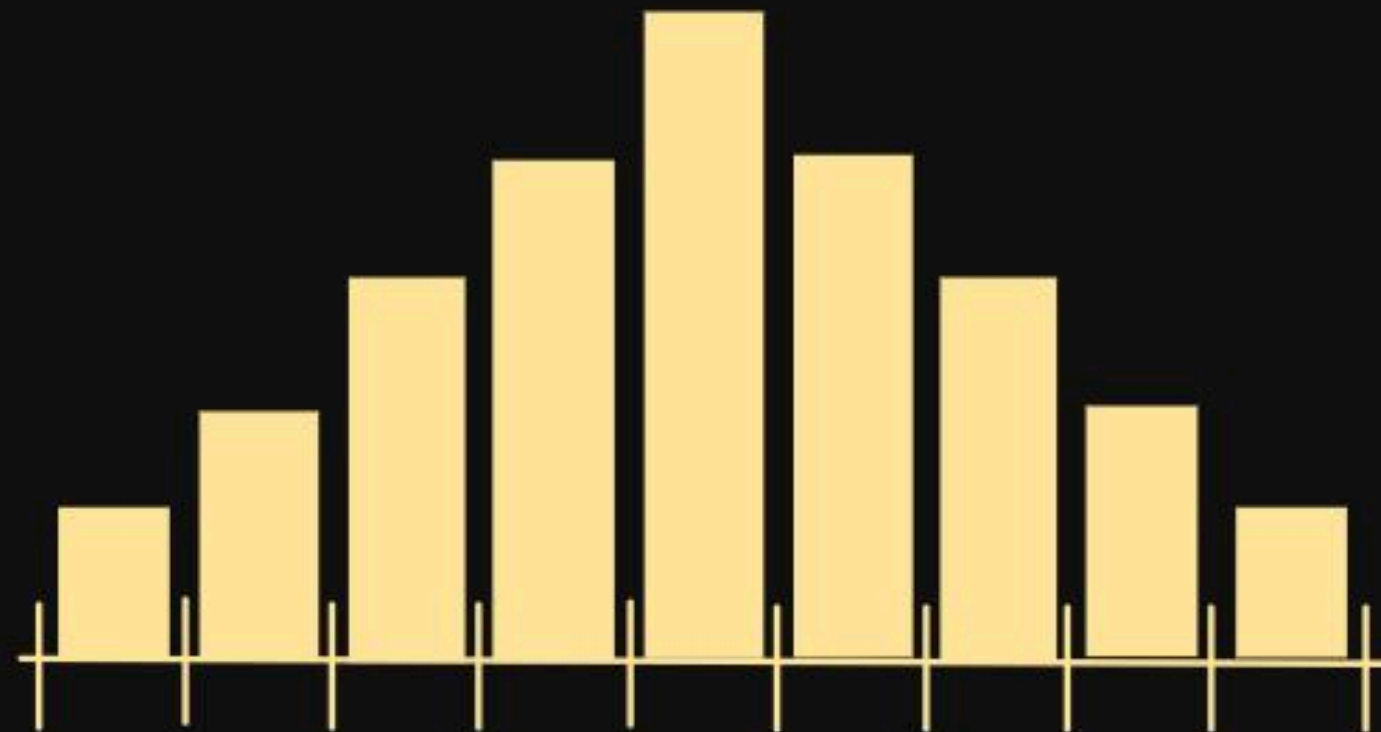
So instead of looking at bars in a histogram, we draw a smooth line/curve over it to see the pattern clearly.

In simple words, A **density curve is just a smooth picture of data distribution** that helps us see the pattern of data clearly.

Density Curve

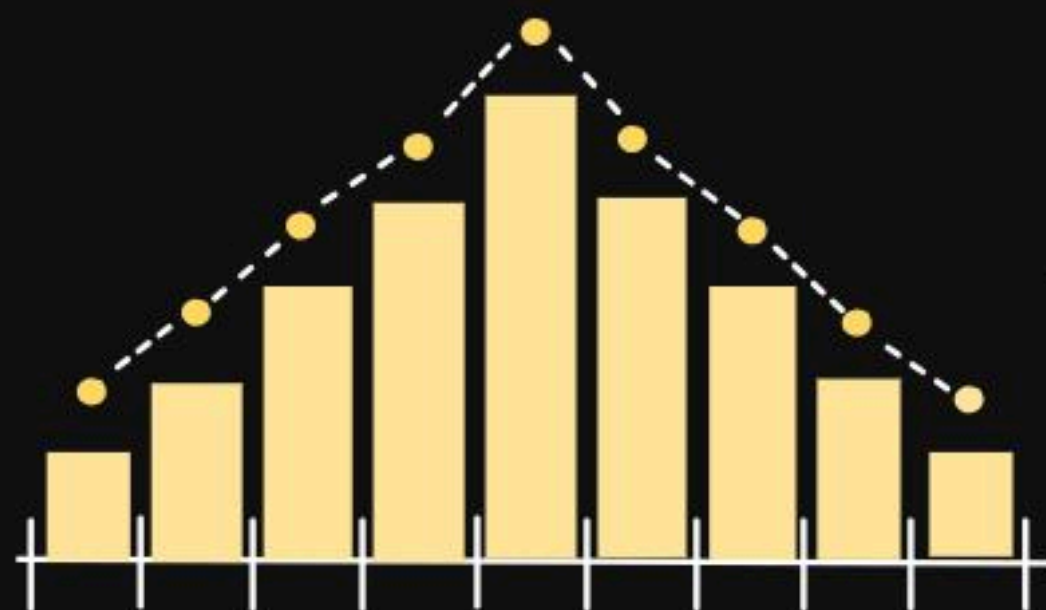
Let's discover the steps to create a density curve now

To understand a density curve should first know about **histogram**, which we have discussed previously.

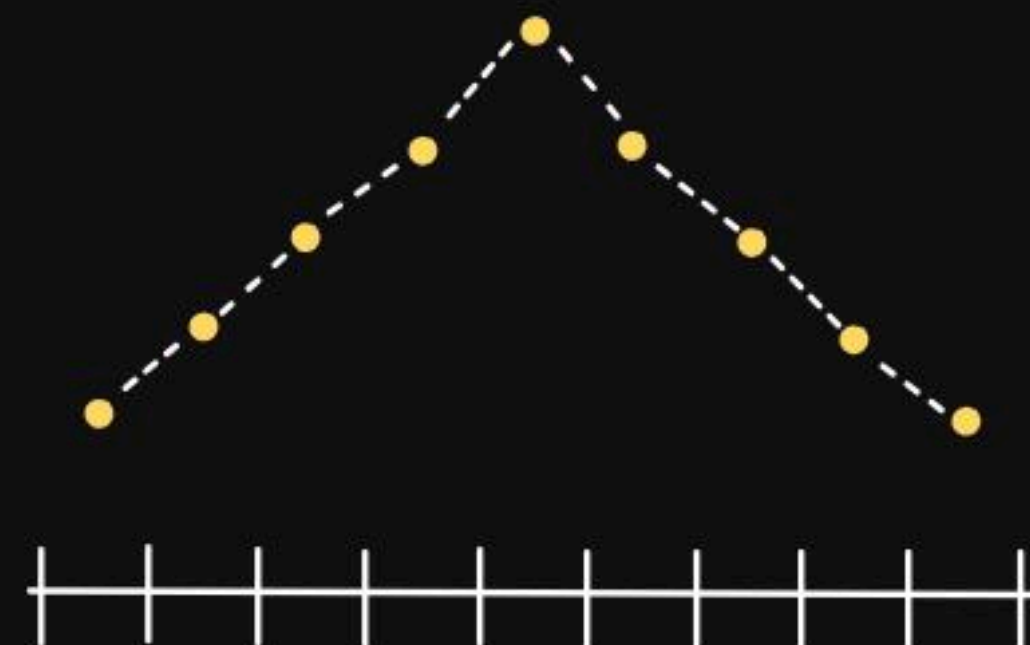


Density Curve

Now we just have to plot points over the bars and connect them kind of making a **line graph**.

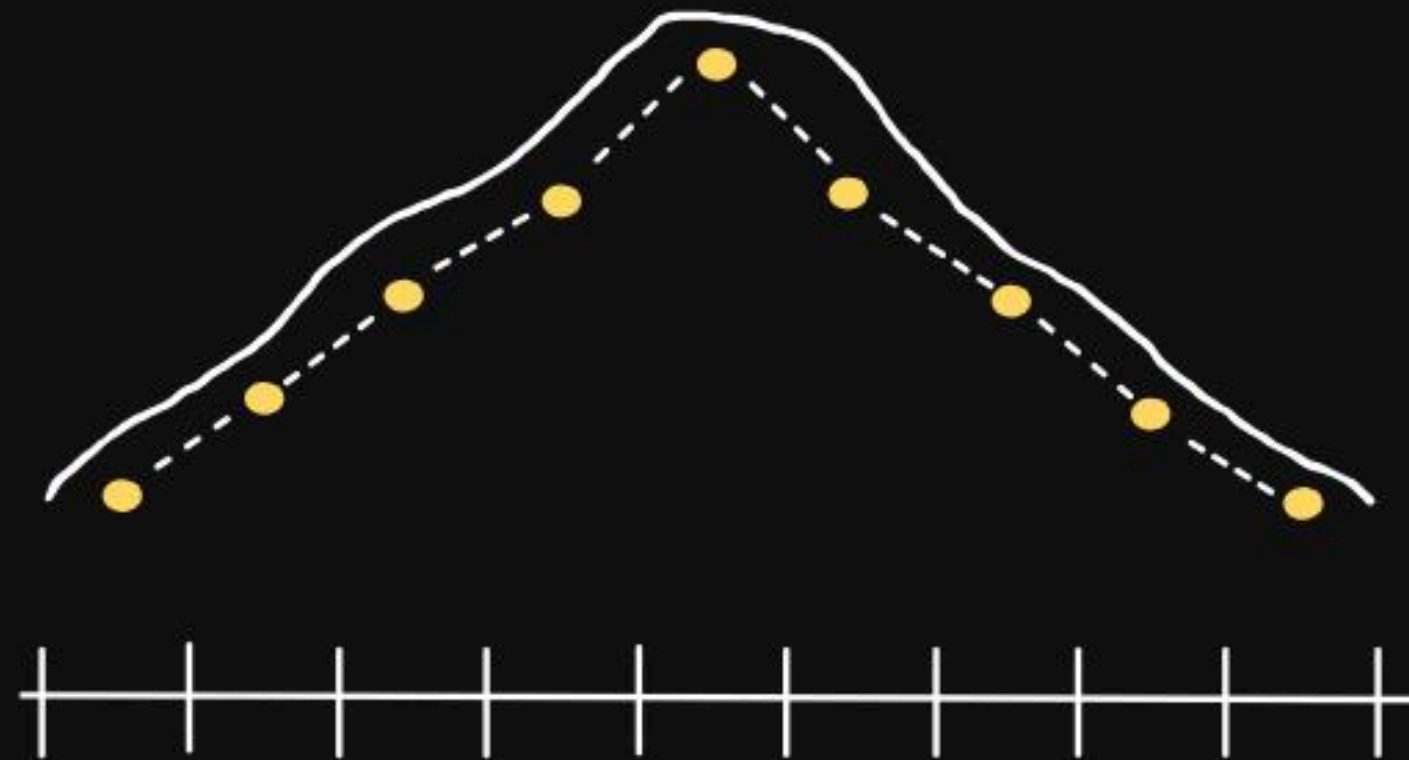


After this now we just have to remove the bars and we will get a frequency polygon.



Density Curve

Now if we **smooth the lines** instead of keeping them sharp, we get a **density curve**.



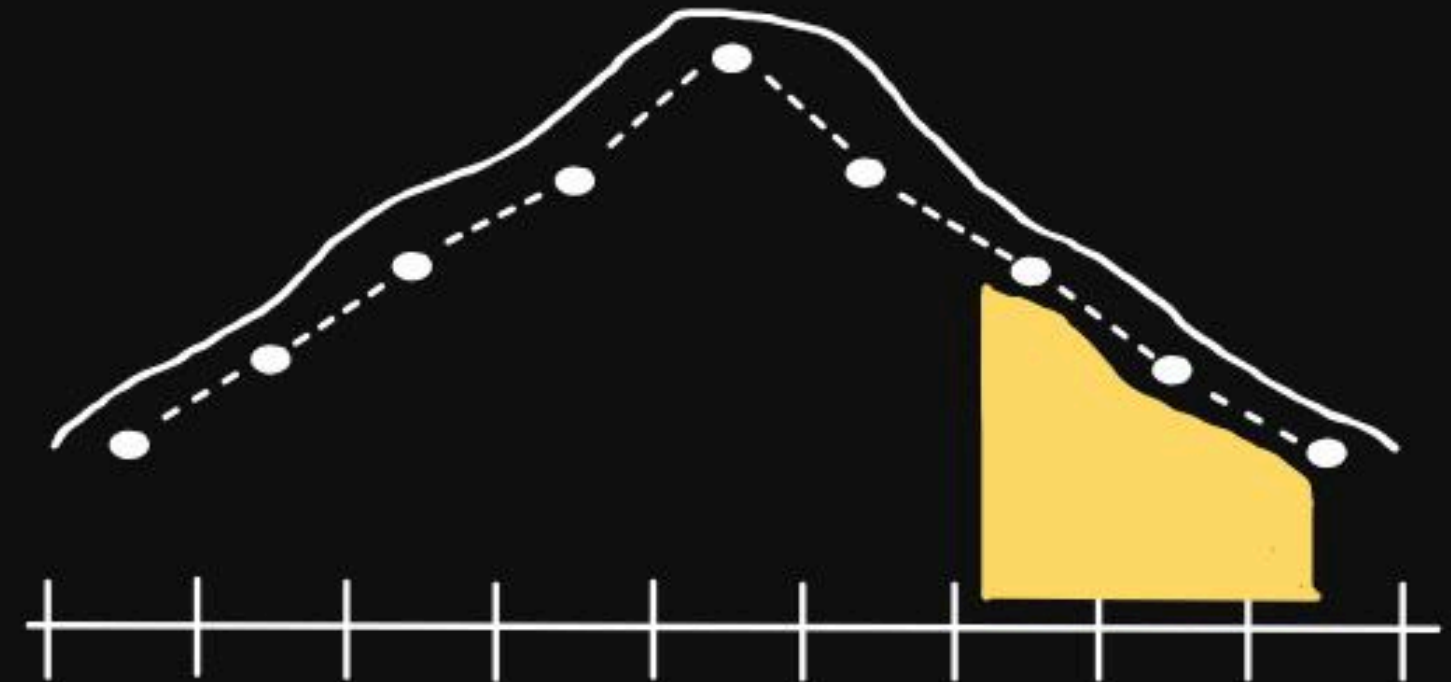
Density Curve

Now this density curve tells us a lot of things.

For example, the total area under the curve is always 100%.

So if we want to know the percentage of data that lies in any range, we can just look at the area of the curve above that range.

That area directly gives us the percentage of our data in that interval.



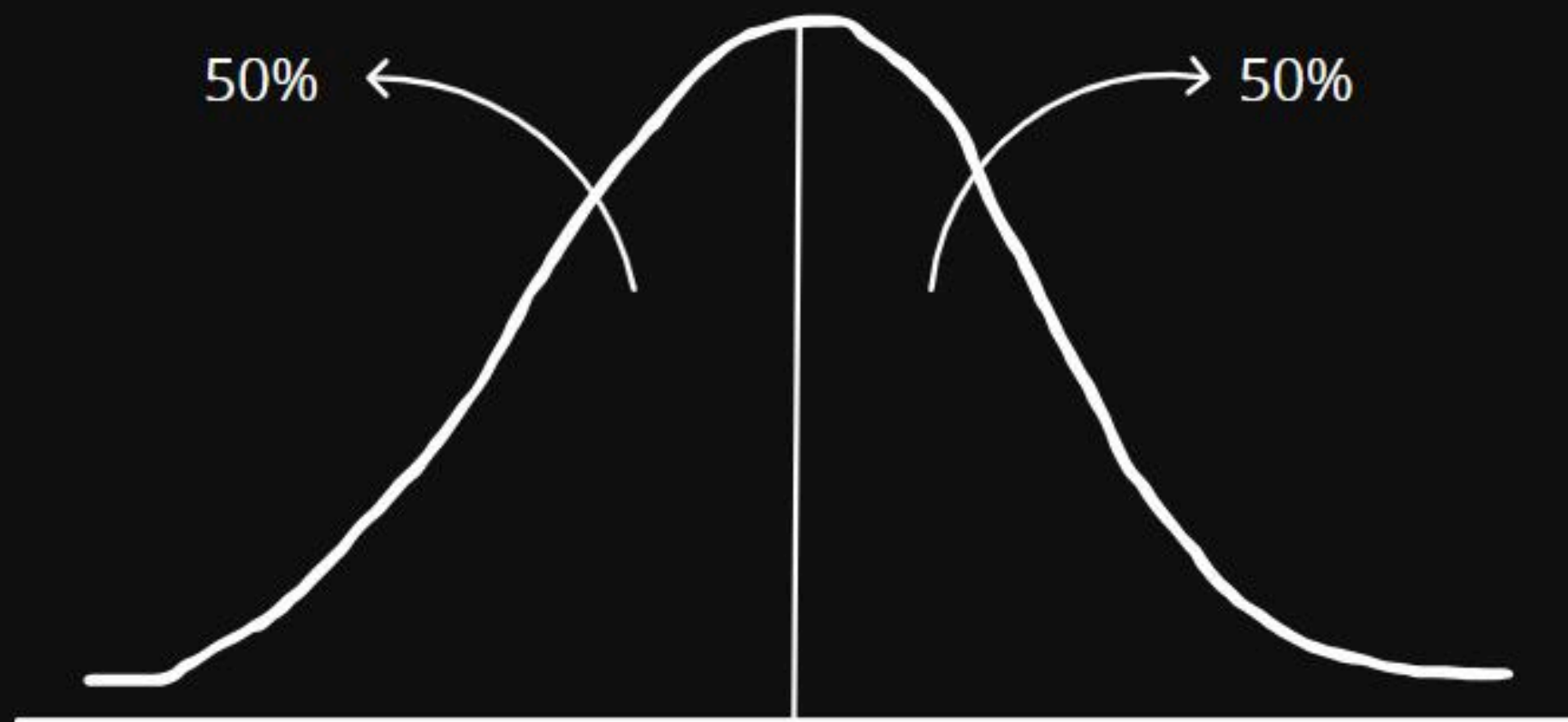
EMPIRICAL RULE

Data Distribution

Empirical Rule

Now we have already seen the normal distribution,
and the area under the normal distribution is always 100% or 1.

The mean is in center and 50% data is on left and 50% data is
on right.

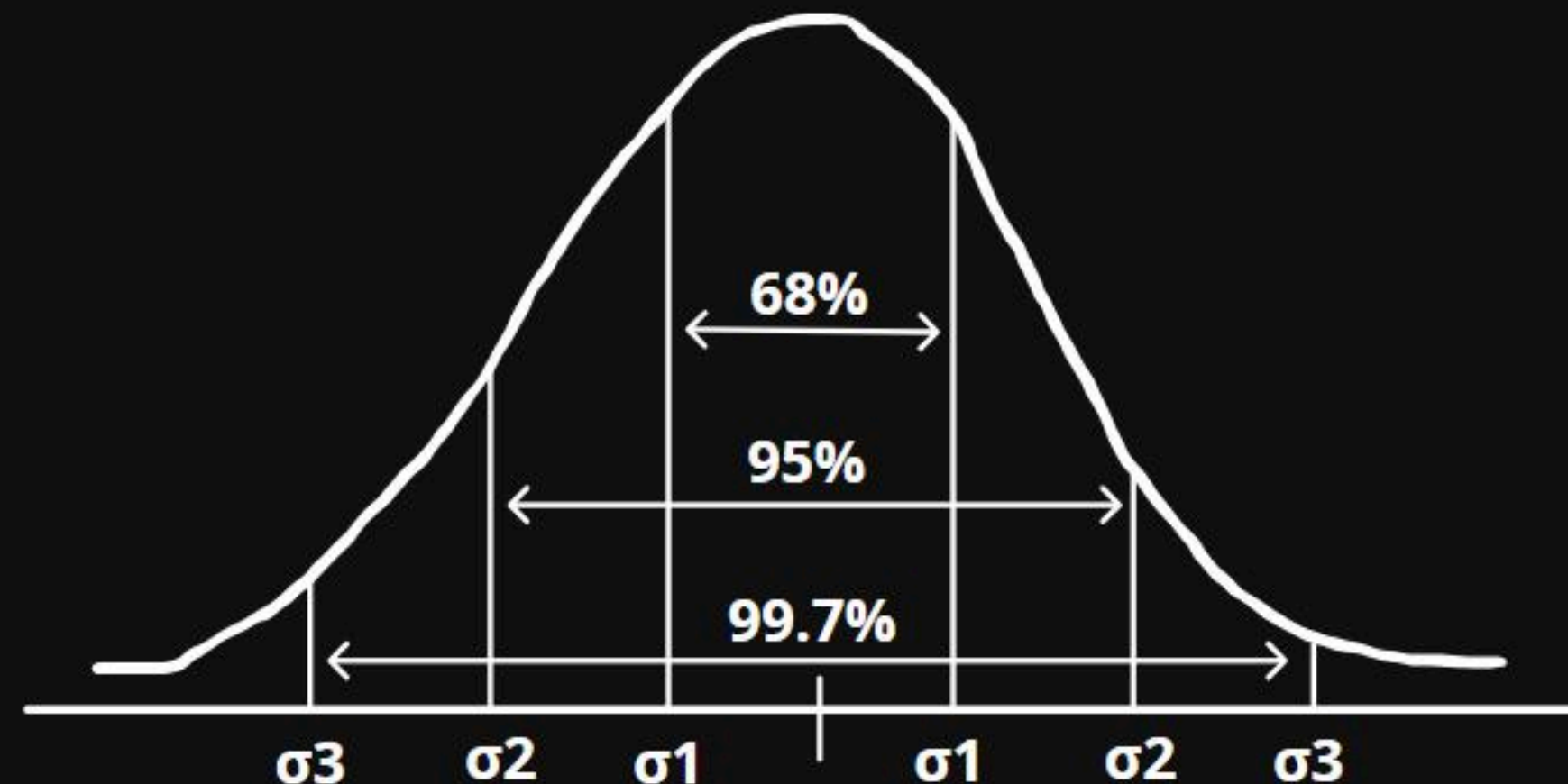


Empirical Rule

In a **normal distribution (bell curve)**:

- About **68%** of the data lies within **1 standard deviation** of the mean.
- About **95%** lies within **2 standard deviations**.
- About **99.7%** lies within **3 standard deviations**.

These lines are divided using standard deviation points.



Empirical Rule

Now lets see an example with a consistent data,

1,2,3,4,5,6,7,8,9

Now here we will find the,

Mean = 5

Standard Deviation = 2.73

